

Flexibility versus Interpretability in High-Dimensional Multiclass Cancer Classification from RNA-Seq Data

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Regression Techniques – Project Proposal (Revised)

1 Objective

We study how predictive performance, model flexibility, feature-selection stability, interpretability, and computational cost trade off when classifying tumor type from extremely high-dimensional gene-expression data. The data are the UCI *gene expression cancer RNA-Seq* set (Fiorini, 2016): $n = 801$ patient samples, $p = 20,531$ continuous RNA-Seq (Illumina HiSeq) features, no missing values, and five tumor classes (BRCA, KIRC, COAD, LUAD, PRAD). The repository lists the task as classification/clustering, so we frame the project as multiclass classification. This replaces our earlier regression formulation, which required a manufactured response.

We compare a progression of models, ordered from simple and interpretable to flexible and opaque:

Logistic $\rightarrow$ Lasso $\rightarrow$ Elastic Net $\rightarrow$ XGBoost $\rightarrow$ Sparse Neural Network.

The goal is not to find the highest accuracy. It is to measure how much predictive gain, if any, we get for each step up in complexity and loss of interpretability. The sparsity, p/n , correlated-predictor, and stability themes of the original proposal are retained.

2 Research Questions

- Q1. Performance.** How does classification performance vary across linear, regularized, tree-ensemble, and neural models in the $p \gg n$ regime?
- Q2. Flexibility.** Does added flexibility improve classification, and how does this depend on the number of genes p and the training-set size n ?
- Q3. Interpretability.** How do the number and nature of selected genes change from sparse linear models to nonlinear and neural models?
- Q4. Stability.** Are selected genes stable under resampling, particularly given strongly correlated expression features? Does Elastic Net select more stably than Lasso?
- Q5. Computational trade-off.** What predictive improvement, if any, is obtained per unit increase in computational cost?

We do not presuppose an outcome. If the neural network is only marginally better than Elastic Net, that is a valid and useful finding.

3 Dataset

- **Size and labels:** 801 samples $\times$ 20,531 genes; class counts BRCA 300, KIRC 146, LUAD 141, PRAD 136, COAD 78 (the imbalance ratio is about 3.8, so all splits are stratified).
- **Scale:** the values are already processed (non-integer, many zeros). We will check whether they are already log-scaled before applying any further transform, and will not apply raw-count methods.
- **Anonymized genes:** features are named `gene_0`–`gene_20530`. Selected genes cannot be biologically annotated, and we treat this as a limitation. Comparisons are among models, not against known biology.

3.1 Exploratory analysis

The EDA answers only questions that affect modelling decisions:

- class balance, and the distribution and zero-inflation of expression values;
- the number of near-constant or mostly-zero genes under candidate filters;
- gene–gene correlation structure (share of pairs with $|\rho| > 0.5, 0.8$, and how crowded a random p -gene subset is at each p), which motivates Elastic Net and the stability analysis;
- PCA (variance explained, effective dimensionality, class separation, outlying samples), with t-SNE or UMAP as an optional visual only.

4 Models

All penalized and neural models use standardized inputs. Standardization is fit on training folds only.

Model	Flexibility	Interpretability	Feature selection
Multinomial logistic	Low	Very high	None
Lasso logistic	Low	Very high	Exact zeros (nonzero coefficients)
Elastic Net logistic	Low–moderate	High	Exact zeros, grouped correlated genes
XGBoost	High	Moderate–low	Importance, permutation, SHAP
Sparse neural net-work	Very high	Low	Group Lasso on input columns

Multinomial logistic regression. $P(Y = k \mid x) = \exp(\beta_k^\top x) / \sum_j \exp(\beta_j^\top x)$. Unpenalized maximum likelihood is not identifiable when $p \geq n$ (and perfectly separable classes make it unstable even below that). We therefore fit it only at $p < n$, or on the top principal components computed within each training fold, and treat it as a conceptual linear baseline.

Lasso logistic regression. Minimize $-\ell(\beta) + \lambda \sum_j |\beta_j|$. It yields a small gene signature together with predictions, and is one of the main models.

Elastic Net logistic regression. Minimize $-\ell(\beta) + \lambda [\alpha \|\beta\|_1 + \frac{1-\alpha}{2} \|\beta\|_2^2]$, with the mixing parameter α tuned over a small grid. Its L_2 component tends to keep correlated genes together, so it is the natural test of whether grouping improves selection stability (Q4).

XGBoost. A gradient-boosted tree ensemble that can capture nonlinear effects and interactions among genes. Tuned parameters: tree depth, learning rate, number of trees (with early stopping on inner validation data), and row/column subsampling. Interpretability is recovered through gain importance, permutation importance, and SHAP values.

Sparse-input neural network. An MLP with one or two hidden layers and softmax output, with an input-level Group Lasso penalty $\lambda \sum_{j=1}^p \|W_{:,j}^{(1)}\|_2$, trained with proximal gradient updates so that whole input columns are set exactly to zero. If $W_{:,j}^{(1)} = 0$, gene j has been removed from the network. The architecture is kept shallow so that flexibility comes from nonlinearity rather than depth.

5 Experimental Design

5.1 Data splits

We hold out a fixed stratified test set (20%, about 160 samples) that is never used for tuning or selection. The remaining pool (about 641 samples) supplies all training data. Results are averaged over repeated stratified splits (at least 20 repeats for the main comparison), with the same splits used across models so comparisons are paired.

5.2 Preprocessing and leakage control

The pipeline is

expression $\rightarrow$ transformation (if needed) $\rightarrow$ variance filtering $\rightarrow$ standardization $\rightarrow$ model.

Every data-dependent step (variance filter, standardization, PCA, importance ranking, supervised selection, hyperparameter tuning) is fit inside the training folds only. Selecting genes on the full data and then cross-validating would inflate performance and is explicitly avoided.

5.3 Varying dimension p

Genes are ranked by training-set variance and the top p are used, with

$$p \in \{50, 100, 500, 1000, 5000, 20531\}.$$

Relative to the training pool ($n \approx 641$), p/n ranges from about 0.08 to 32. The largest value uses all genes with no filtering and is run for the penalized and neural models, and for XGBoost if computation allows.

5.4 Varying training size n

At a fixed p (and repeated across a small set of p values), train on $n \in \{50, 100, 200, 400, 600\}$ stratified subsamples of the training pool and evaluate on the fixed test set. This shows how much data each model needs for reliable classification.

5.5 Hyperparameter tuning

Penalties and other hyperparameters are tuned by cross-validation within the training data (nested where feasible). For Lasso and Elastic Net we report both the minimum-error and 1-SE choices, since the 1-SE rule gives sparser signatures. Neural width and Group Lasso strength are chosen from validation curves.

6 Evaluation

Predictive metrics. Accuracy, macro-F1, and log-loss overall; precision, recall, and F1 per class (BRCA, KIRC, COAD, LUAD, PRAD); and confusion matrices showing which tumor types are confused. Macro-F1 and per-class recall matter because COAD is the smallest class. We expect accuracy to be near its ceiling at large n and p , so the informative differences are likely to appear at small n , small p , and in log-loss and sparsity, and we design the grids accordingly.

Interpretability and selection. For Lasso and Elastic Net, the selected set is the nonzero coefficients. For XGBoost and the neural network there are no exact zeros, so we use the top- k genes by SHAP/permutation importance and by input-weight norm respectively, with k matched to the median size of the Lasso signature so that comparisons of selected sets are fair. We then compare $S_{\text{Lasso}}, S_{\text{EN}}, S_{\text{XGB}}, S_{\text{NN}}$ (overlap and agreement).

Stability. Over $B \approx 50$ bootstrap resamples, we compute the Jaccard similarity $J(A, B) = |A \cap B|/|A \cup B|$ between selected sets, and each gene’s selection frequency (the share of resamples in which it is selected). We compare prediction stability against selection stability, since a model can predict well while choosing different genes each time.

Computational cost. Wall-clock training time, number of parameters, number of selected genes, and epochs to convergence for the neural network, all as functions of p and n .

Summary table (to be filled after experiments).

Model	Accuracy	Macro-F1	# Genes	Stability	Time	Interpretability
Logistic						Very high
Lasso						Very high
Elastic Net						High
XGBoost						Medium
Sparse NN						Low

Central figures. (i) Performance versus model flexibility; (ii) number of selected genes versus performance; (iii) performance curves over p and over n .

7 Planned Report Structure

Introduction; Dataset; Exploratory analysis; Statistical methodology; Experimental design; Predictive performance; Feature selection; Stability; Flexibility versus interpretability; Computational cost; Discussion (when is a sparse linear model sufficient, when does nonlinearity help, what is lost in interpretability, how stable are the signatures); Conclusion.

8 Work Plan and Division of Labour (proposed)

Member	Primary responsibility	Also contributes
Rajveer Vora	Data pipeline, EDA, Lasso/Elastic Net models, splits and leakage control	Stability analysis code, report integration
Amartya Amritanshu	XGBoost, SHAP/permutation importance, feature-set comparison	Timing experiments, figures
P. Sanjith Reddy	Sparse neural network (Group Lasso, proximal training)	Varying- n experiment, discussion section

This split is provisional and can be adjusted.

Phase	Content
1	Data loading, preprocessing, EDA, split and CV framework
2	Logistic, Lasso, Elastic Net pipelines; first results across p
3	XGBoost and sparse neural network implementations and tuning
4	Full p and n grids; repeated splits; timing
5	Selection comparison and bootstrap stability analysis
6	Analysis, figures, report and discussion

9 Deliverables

1. A reproducible code base (fixed seeds, shared splits) covering preprocessing through all experiments.
2. Predictive-performance results across models, p , and n , with per-class metrics and confusion matrices.
3. Feature-selection and stability results, including cross-model gene agreement.
4. Computational-cost comparison.
5. A written report answering Q1–Q5, with a discussion of when the extra flexibility pays for itself and when a sparse linear model suffices.

10 Limitations and Risks

- **Ceiling effects.** Tumor types may be nearly perfectly separable, leaving little room for model differences. The n - and p -grids, log-loss, and sparsity metrics are included to expose differences that accuracy alone would hide.
- **Anonymized genes.** Selected genes cannot be biologically validated, and the study makes no biomarker or diagnostic claims. A single cohort cannot support them without independent validation.
- **Comparing selection across model types.** Exact-zero selection and importance-ranked selection are not the same notion. Matching set size at k mitigates this but does not remove it.

- **Cost of the full grid.** The $p = 20531$ setting with repeated splits and bootstraps is expensive for XGBoost and the neural network, so the number of repeats may be reduced there and this will be reported.
- **Non-convex training.** Neural training depends on initialization, so multiple random starts are used.

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